

BRAC UNIVERSITY
Department of Computer Science and Engineering

Set:A

Examination: Mid Term
Duration: 1 Hour 30 Minutes

Semester: Summer 2026
Full Marks: 25

CSE 340: Computer Architecture

Answer the following questions. Show calculations where required. Figures in the right margin indicate marks. Understanding the question is part of the examination.

Name:	ID:	Section:
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1. a. Two laptops, Titan and Falcon, run the same ISA. Titan has a clock rate of 2.8 GHz and CPIs for instruction classes A, B, C, D are 7, 4, 2, 3. Falcon has a clock rate of 3.5 GHz and CPIs for instruction classes A, B, C, D are 5, 3, 2, 2.
- 1 A program with 3.0×10^6 instructions is made up of:

- 50% class A
- 20% class B
- 20% class C
- 10% class D

i. **Calculate** the average CPI for both Titan and Falcon. [1+1]

ii. **Calculate** the execution time for both systems. [2+2]

iii. If a reference machine takes 120 ms to run the same program, **calculate** the SPECRatio for Falcon. [2]

- b. A company stores all of its critical customer data on a single hard drive with no backups. One day the hard drive fails, and all the data is permanently lost, causing major disruption to the business. To prevent this from happening again, the company sets up a new system where the same data is automatically copied and stored on two additional separate hard drives. If one drive fails, the data can still be recovered from the other two. [1+1]

Which of the seven great ideas of Computer Architecture is this scenario supporting/violating (in its initial setup vs. after the fix)? **Explain** with proper identification and justification.

2. Assume, $Y = 1.1011 \times 2^{-2}$, $X = 1.0000 \times 2^3$

- C a. **Determine** the value of P if $P = X - Y$ using floating point subtraction. **Show** detailed calculation. [2]
- O b. In a custom IEEE754 floating point representation where there are 3 bits assigned for the exponent field, **check** if the answer from 2.a is going to overflow/underflow. [1.5]
- 1 c. We want to convert Y into a custom IEEE754 floating point representation, **what** would be the minimum number of bits to represent the sign, exponent and fraction value? [1.5]

3. a. **Determine** whether the following statements are true or false, if the statement is **false**, then write the correct answer. [1+1]
- C
O
2
- i. Using SB-format instructions we can only perform conditional jumps.
- ii. S-Type instruction format uses one destination register and one source register.

- b. **Explain** how the LUI and JALR instructions work together to construct a 32-bit absolute target address and perform a long jump in RISC-V. [2]

- c. Consider the following partially incomplete RISC-V program. An instruction is missing at address **4056**. Their corresponding machine codes are provided. [2]

Address(Decimal)	Instruction	Machine code
4032	addi x7, x0, 100	
4036	andi x9, x18, 60	
4040	bne x9,x7, label2	
4044	ld x5, 0(x16)	
4048	addi x5, x5, -1	
4052	sh x5, 0(x16)	
4056	_____	0xFE92BA4D
4060	lh x5, 2(x16)	
4064	srli x5, x5, 1	
label 2:4068	sh x5, 2(x16)	
4072		

Assume that the missing instruction at address 4056 is a **branch** instruction. Opcode for any branch instruction = 1001101. Funct3 for beq = 010, bne = 001, bge = 000, blt = 011.

Decode the machine code into its corresponding assembly instruction.

Note: Use "**label4**" as the label for the branch instruction and **write** the final answer in your script.

- d. **Convert** the following high level language into its corresponding assemble language: [4]

<pre>sum = calc(1,10); a[5] = sum*127</pre>	<pre>int calc(int start, int size) { int result = 0; for (int i = start; i < size; i++) { if (a[i] == 0) { result += a[i]; } else { result -= a[i]; } } return result; }</pre>	You must consider the following: • a is a double word array; • Base address of array a is in register x15; • variable sum is associated with register x22; • pass the argument using register x13, x14; • variable result also uses register x22 and variable i uses x24; • return the value using register x10.
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